



Identifying Risks & Mitigation Plans

No infrastructure project runs perfectly. Your capstone success depends on anticipating **technical, operational, and compliance risks** while building **resilience into your design** from the start.

Why Risk Planning Matters

AI infrastructure is both costly and complex, with failures potentially wasting valuable **budget and time**. Without proper planning, your capstone could face:

- Compliance failures
- Critical data loss
- Extended system downtime



Effective mitigation strategies prevent your capstone from derailing, saving both resources and your grade.

Categories of Risks

Technical

Hardware limitations, software compatibility issues, and scalability challenges that can derail your system architecture.

Operational

Deployment failures, monitoring gaps, and process breakdowns that affect day-to-day functioning.

Compliance & Security

Data leaks, IAM misconfigurations, and regulatory violations that expose sensitive information.

Financial

Cost overruns, hidden cloud charges, and budget miscalculations that deplete resources.

Each category requires specific strategies to identify, evaluate, and mitigate potential problems.

Common Technical Risks

Primary Risks:

- GPU/TPU unavailability in your selected cloud provider
- Training jobs crashing due to out-of-memory (OOM) errors
- Model convergence issues leading to wasted compute resources

Effective Mitigations:

- Implement spot capacity fallback to on-demand instances
- Use smaller batch sizes and gradient checkpointing techniques
- Deploy early-stopping algorithms with continuous monitoring



Operational Risks



Kubernetes Misconfigurations

Pod failures, resource constraints, and deployment issues can bring your entire pipeline down.



Logging Gaps

Without proper observability, you'll be blind to performance drift and emerging problems.



CI/CD Failures

Broken deployment pipelines can prevent updates and fixes from reaching production.

✔ Mitigation Strategies

- Use templates (Helm, Terraform) to standardize deployments
- Implement canary rollouts before full production releases
- Build redundancy into your monitoring systems

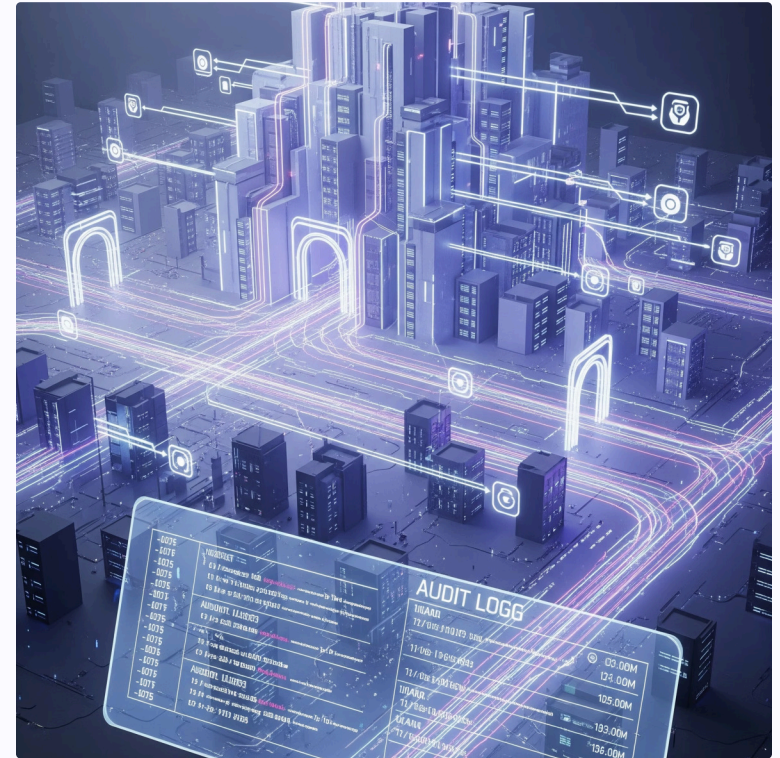
Compliance & Security Risks

Primary Security Concerns:

- Non-compliant storage of sensitive training data
- IAM misconfigurations leading to unauthorized access
- Lack of encryption resulting in data leaks during transit

Effective Mitigations:

- Encrypt all data (both at rest and in transit)
- Apply RBAC with least-privilege access policies
- Implement immutable logs and comprehensive audit trails



Financial Risks

Underestimated GPU Training Costs

Even small models can incur thousands in computing charges when hyperparameter tuning is extensive.

Cross-Region Data Transfer Charges

Egress fees can add up quickly when moving large datasets between regions or providers.

Idle GPU Waste

Overprovisioned clusters lead to expensive compute resources sitting unused.

Financial Safeguards:

Use spot/preemptible instances when safe, co-locate data with compute resources, and implement auto-scaling with strict budget alerts.



Risk Prioritization Matrix

Risk	Likelihood	Impact	Mitigation
GPU unavailability	High	High	Fallback to on-demand instances
Cost overruns	Medium	High	FinOps dashboards + alerts
Data compliance	Low	High	Encryption + IAM controls
Drift undetected	Medium	Medium	Prometheus + EvidentlyAI

Prioritize risks by combining likelihood and impact scores to focus your mitigation efforts where they matter most. High-High risks require immediate attention and robust planning.

Case Study: Healthcare AI Risk Planning

Risk: HIPAA Compliance Breach

Mitigation: Implemented federated learning architecture that keeps patient data local, with de-identification pipelines for any data that must be centralized.

Risk: Model Drift on New Imaging Devices

Mitigation: Developed automated drift detection pipeline with statistical monitoring and retraining triggers when performance metrics decline.

The healthcare team protected both patient privacy and model performance by anticipating regulatory and technical challenges.



Case Study: Startup NLP Service

Risk: Spot GPUs
Preempted Mid-
Training

Mitigation: Implemented automatic checkpointing every 30 minutes to allow seamless training resumption after preemption, reducing wasted compute by 87%.

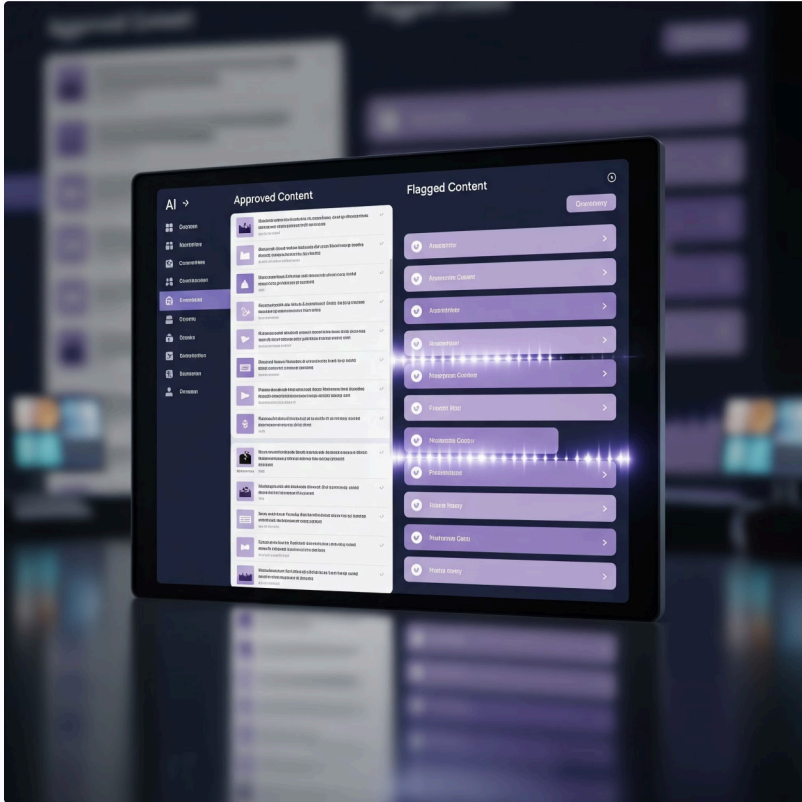
This startup saved over \$120,000 in its first year through these proactive measures while maintaining high availability.

Risk: Runaway Cloud
Billing

Mitigation: Set up granular billing alerts at 50%, 75%, and 90% of budget with automatic usage quotas that prevent new job submissions when thresholds are reached.



Case Study: GenAI Content Service



Risk: Unsafe Generations

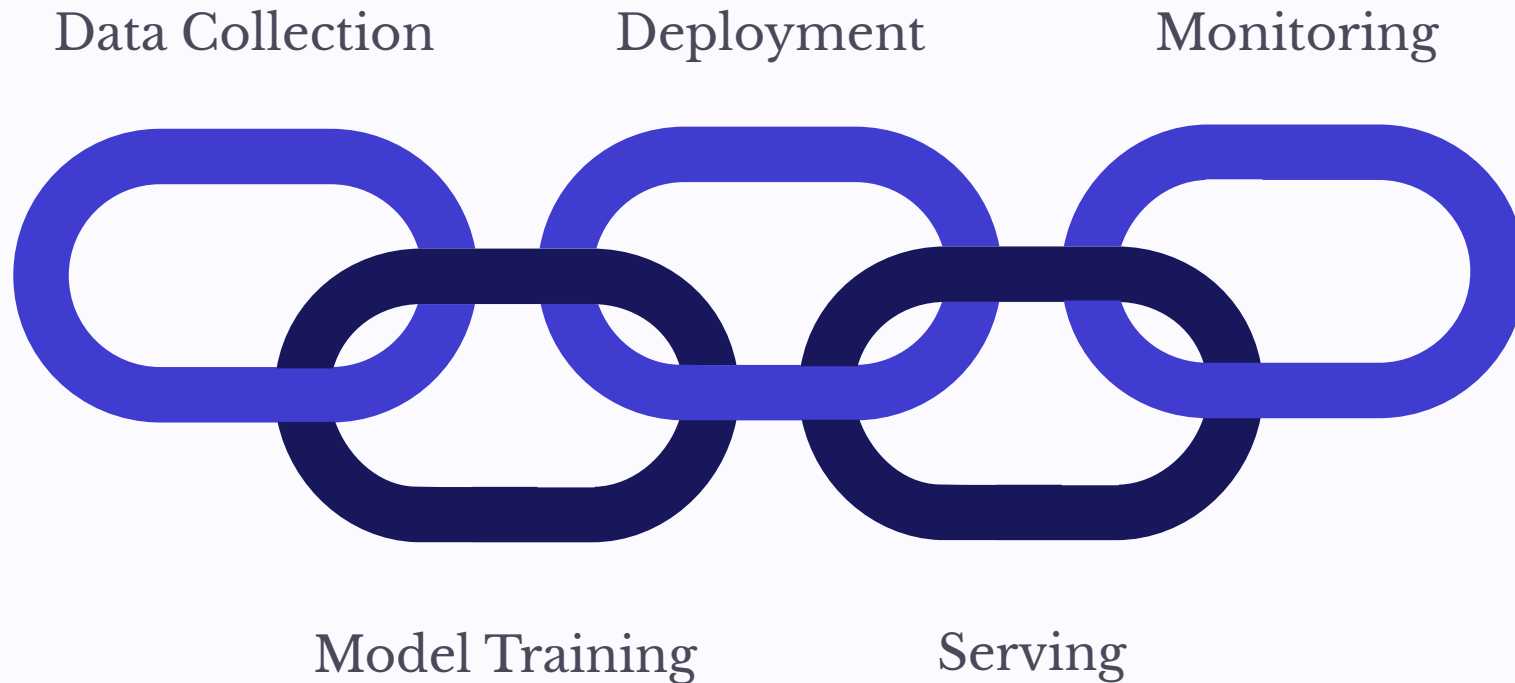
Mitigation: Implemented multi-stage content filtering pipeline and RLHF alignment techniques to prevent reputational damage.

Risk: Scaling Inference Costs

Mitigation: Built response caching system for frequent queries and distilled the large model into a smaller, more efficient version for common tasks.

These measures reduced inference costs by 62% while maintaining generation quality and safety standards.

AI Infrastructure Risk Lifecycle



Resilient infrastructure = risks mapped and planned at every stage.

Effective risk management follows your system through its entire lifecycle, with specific mitigation strategies tailored to each phase.

Key Lessons

01

Comprehensive Coverage

Risks span **technical, operational, compliance, and financial** domains. Missing any category leaves your project vulnerable.

03

Communication Tool

A well-constructed risk matrix serves as both planning document and stakeholder communication tool.

02

Proactive Approach

Mitigation belongs in early design phases, not as emergency firefighting. Build safeguards into your architecture from day one.

04

Proposal Requirement

Your capstone proposal **must include a dedicated risk section** demonstrating thoughtful analysis and planning.

Summary



Your Risk Management Roadmap

- Identify risks across all infrastructure layers
- Prioritize based on likelihood and impact
- Document specific mitigation strategies
- Implement controls before problems arise

This approach not only builds **technical resilience** into your capstone, but also demonstrates the **professional maturity** that employers seek in ML engineers.



Discussion Prompt

If you had to choose one area to over-invest in risk mitigation — **technical stability**, **compliance**, or **cost control** — which would you pick, and why?

Consider your specific application domain, team resources, and the consequences of failure in each area. There's no universally correct answer, but your reasoning reveals your priorities.